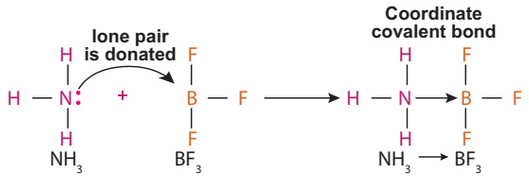
molecule. Here the molecule ammonia (NH_3) gives a lone pair of electrons to Boron tri fluoride (BF_3) molecule which is electron deficient. Thus, a coordinate covalent bond is formed between NH_3 (donor molecule) and BF_3 (acceptor molecule) and is represented by $NH_3 \rightarrow BF_3$.



Characteristics of coordinate covalent compounds

The compounds containing coordinate covalent bonds are called coordinate compounds.

Physical state: These compounds exist as gases, liquids or solids.

Electrical conductivity: Like covalent compounds, coordinate compounds also do not contain charged particles (ions), so they are bad conductors of electricity.

Melting point: These compounds have melting and boiling points higher than those of purely covalent compounds but lower than those of purely ionic compounds.

Solubility: Insoluble in polar solvents like water but are soluble in non-polar solvents like benzene, CCl_4, and toluene.

Reactions: Coordinate covalent compounds undergo molecular reactions which are slow.

13.4 Oxidation, Reduction and Redox reactions

When an apple is cut and left for sometimes, its surface turns brown. Similarly, iron bolts and nuts in metallic structures get rusted. Do you know why these are happening? It is because of a reaction called oxidation.



Oxidation: The chemical reaction which involves addition of oxygen or removal of hydrogen or loss of electrons is called oxidation.

$2\ Mg + O_2 \rightarrow 2\ MgO$ (addition of oxygen)
$CaH_2 \rightarrow Ca + H_2$ (removal of hydrogen)
$Fe^{2+} \rightarrow Fe^{3+} + e^-$ (loss of electron)

Reduction: The chemical reaction which involves addition of hydrogen or removal of oxygen or gain of electrons is called reduction.

$2\ Na + H_2 \rightarrow 2\ NaH$ (addition of hydrogen)
$CuO + H_2 \rightarrow Cu + H_2O$ (removal of oxygen)
$Fe^{3+} + e^- \rightarrow Fe^{2+}$ (gain of electron)

Redox reactions: Generally, the oxidation and reduction occurs in the same reaction (simultaneously). If one reactant gets oxidised, the other gets reduced. Such reactions are called oxidation-reduction reactions or Redox reactions.

$$2\ PbO + C \rightarrow 2\ Pb + CO_2$$
$$Zn + CuSO_4 \rightarrow Cu + ZnSO_4$$

Oxidation	Addition of oxygen
	Removal of hydrogen
	Loss of electron
Reduction	Removal of oxygen
	Addition of hydrogen
	Gain of electron

Oxidising agents and Reducing agents

Substances which have the ability to oxidise other substances are called oxidising agents.

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